University of Bologna
Master in Digital Humanities and Digital Knowledge

UNIFYING KHMER CULTURAL HERITAGE:

A Semantic Data Model for Cross-Institutional Integration

THESIS DEFENSE

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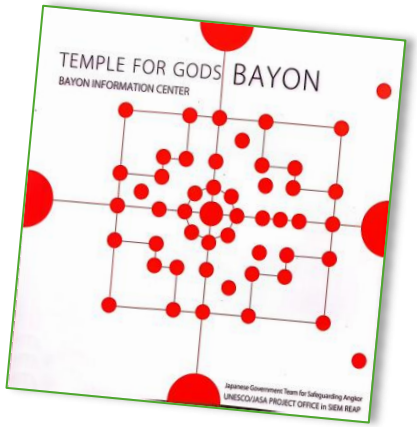
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Co-supervisor: Ph.D. Valentina Pasqual

October 2026

MUSEUM COLLECTIONS



RESEARCH & CONSERVATION REPORTS



Problem: Fragmented heritage data



Temple occupation and the tempo of collapse at Angkor Wat, Cambodia

SCIENTIFIC PUBLICATIONS

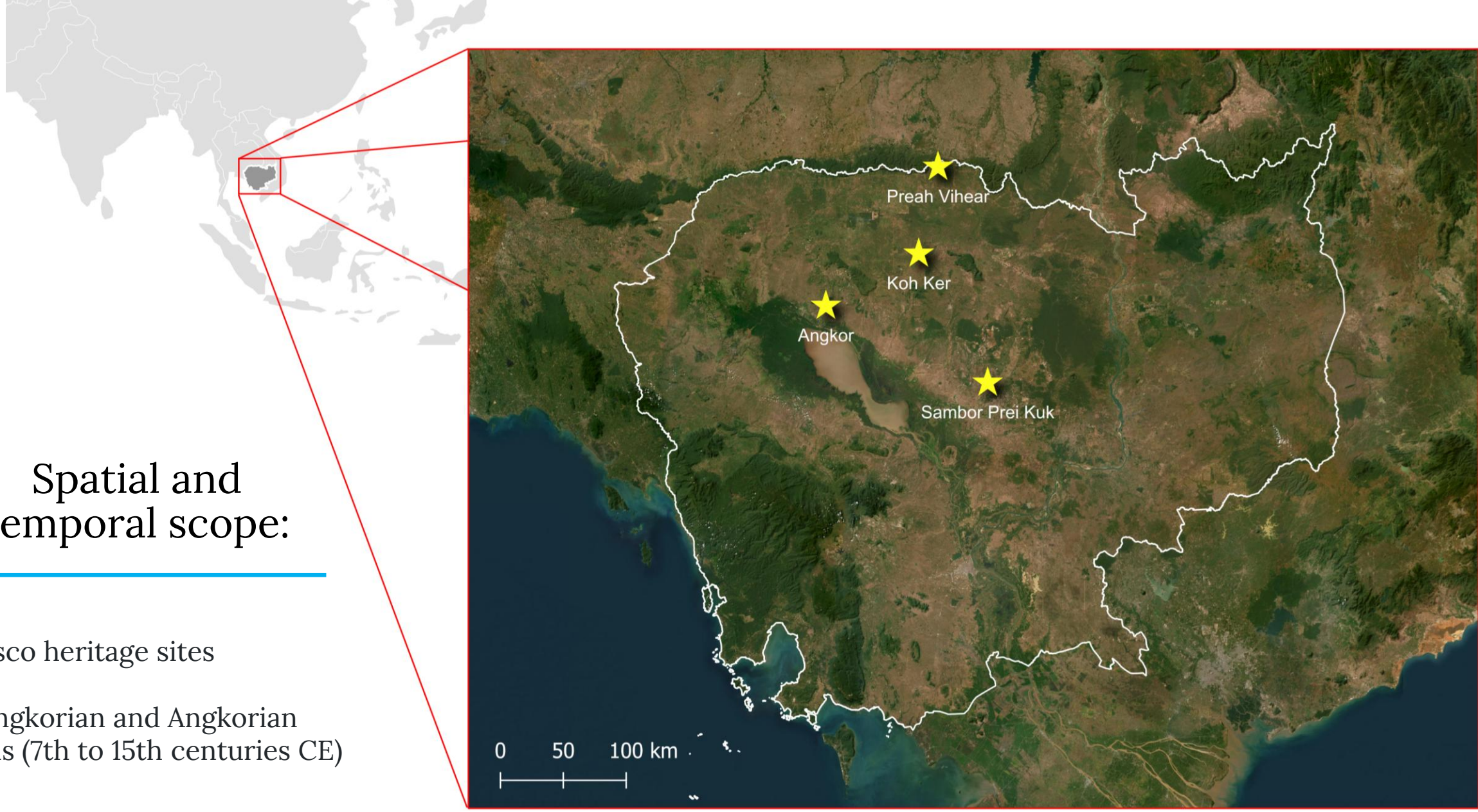
ARCHAEOLOGICAL & GEOSPATIAL DATA



Spatial and temporal scope:

4 Unesco heritage sites

Pre-Angkorian and Angkorian periods (7th to 15th centuries CE)



Source: Data from UNESCO; ESRI Satellite / Created by the author

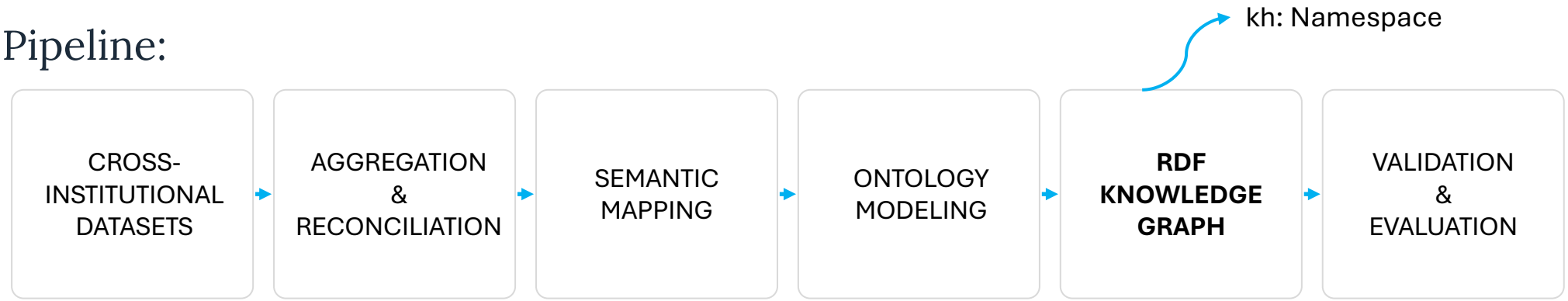
Research Question & Objectives

How can fragmented data be integrated to provide a comprehensive digital representation of the Khmer cultural heritage domain within the Semantic Web?

- O1 Aggregate and harmonize heterogeneous data.
- O2 Align metadata practices through established semantic models.
- O3 Manage temporal, geographical and terminological variation.
- O4 Support relationships and retrieval / visualization methods.
- O5 Contextualize the geographical displacement of objects over time.

Methodology

Pipeline:



DATASET COMPOSITION

Musée Guimet • National Museum of Cambodia • WebOfScience • Scopus • Google Scholar • UNESCO reports • APSARA reports • ESRI Geodata

MAIN SEMANTIC FRAMEWORK

CIDOC CRM

COMPLEMENTARY MODELS

DCTerms • Bibo • GeoSPARQL • FOAF • Schema • RDFS • SKOS

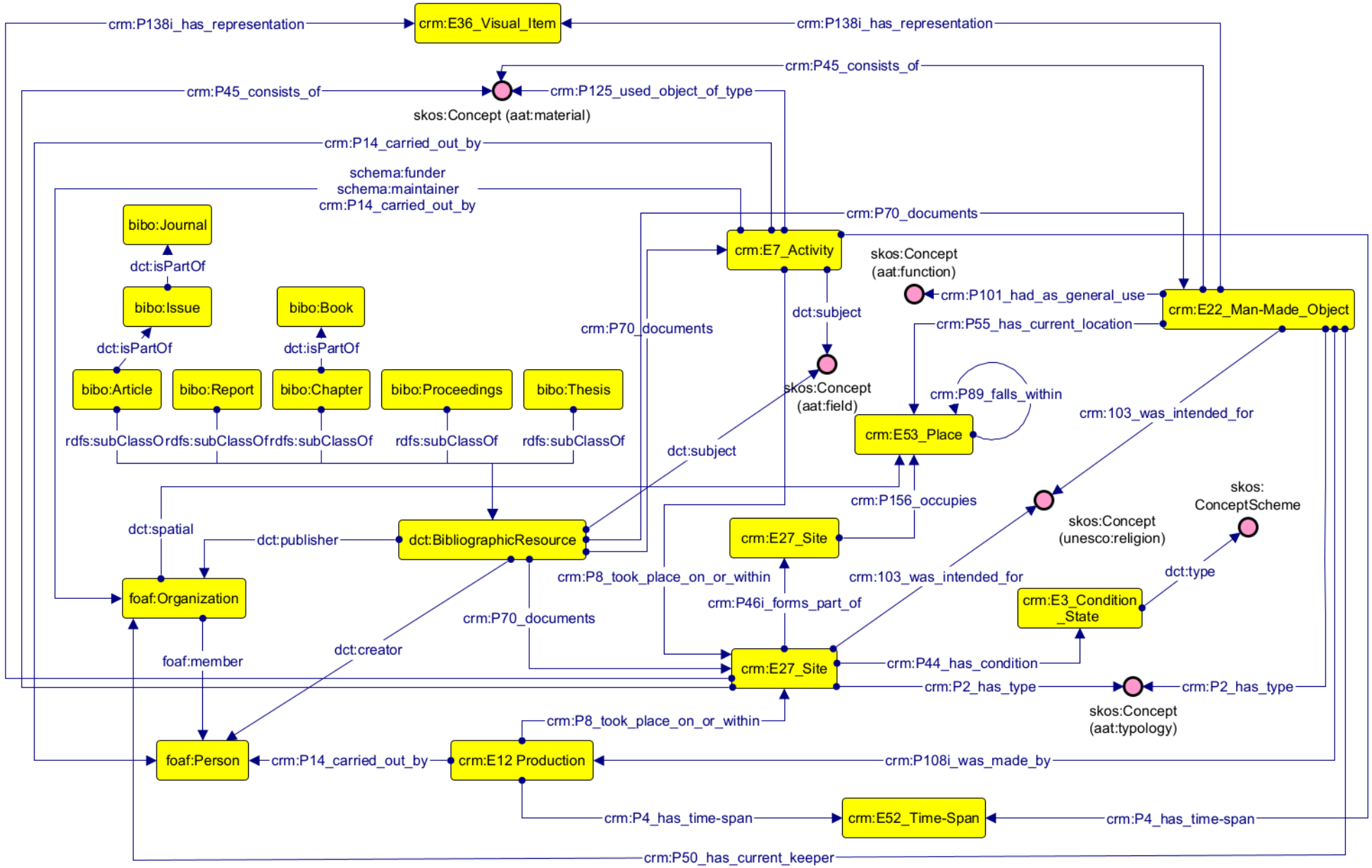
CONTROLLED VOCABULARY

AAT • Wikidata • GeoNames

VALIDATION

Protégé Reasoning + SHACL + Competency questions

Conceptual Model



Results: Knowledge Graph

— GRAPH SCALE

3,272 named individuals

— FOUR INTERCONNECTED AXES

229 objects (`crm:E22_Human-Made Object`)
177 sites (`crm:E27_Site`)
20 conservation projects (`crm:E7_Activity`)
231 Publications (`dct:BibliographicResource`)

— CONTEXTUAL NODES

63 places (`crm:E53_Place`)
392 persons (`foaf:person`)
230 organizations (`foaf:organization`)
402 production events (`crm:E12_Production`)
421 time-spans (`crm:E52_Time-Span`)
17 materials (`skos:Concept`)

Results: Competency questions

Internal and cross-domain retrieval

Q1. Which scientific fields produce most publications related to Khmer heritage?

Q2. Which publications are associated with museum objects?

Q3. Which materials were employed in the construction of temples at Sambor Prei Kuk?

Q4. Which countries host institutions that preserve objects dating back to the ninth century?

Q5. Which institutions produce the largest volume of publications concerning Preah Vihear Temple?

Q6. Which projects focus on constructions commissioned during the reign of Jayavarman IV ?

Discussion

Queries validate retrieval

SPARQL execution shows that intended relationships can be retrieved; it does not prove historical truth.

Modeling is interpretative

More than a strict technical implementation, modeling is an epistemic process.

Interoperability ≠ uniformity

The model connects sources while preserving differences in terminology, granularity and interpretation.

Current limitations

- Sample corpus and limited source availability
- Manual entity reconciliation does not scale easily
- Object custody trajectories remain incomplete
- No visualization interface implemented or evaluated

Further developments

- Expand corpus and institutional coverage
- Semi-automated reconciliation workflows
- Web interface: maps, timelines, networks and provenance views
- Represent competing scholarly interpretations

Conclusion

Technical contribution

A queryable RDF knowledge graph grounded in CIDOC CRM that connects four dimensions of Khmer cultural heritage documentation.

Methodological contribution

Semantic integration as a process that balances structural normalization, granularity, uncertainty, and must be documented beyond technical implementation.

Research contribution

Unifies fragmented Khmer documentation into a semantic framework that enables scholars to navigate across distributed institutional archives and gather connected information.

Thank you • Questions